Modeling and Linearization of the Quanser 5-Bar Planar Manipulator (Simplified 2-Link Model)

Control Systems II coursework

1 System Overview

The Quanser 5-bar manipulator is a planar parallel robot actuated by two DC motors. In this simplified model, we consider only the two actively driven links (Link 1 and Link 2), and neglect the passive links and gravity, assuming motion entirely in the horizontal plane.



Figure 1: System schematic showing the two actuated links (Link 1 and Link 2) and their coordinate definitions in the Quanser frame

The coordinate system follows the official Quanser manual: joint angles θ_0, θ_1 are measured from the vertical axis.

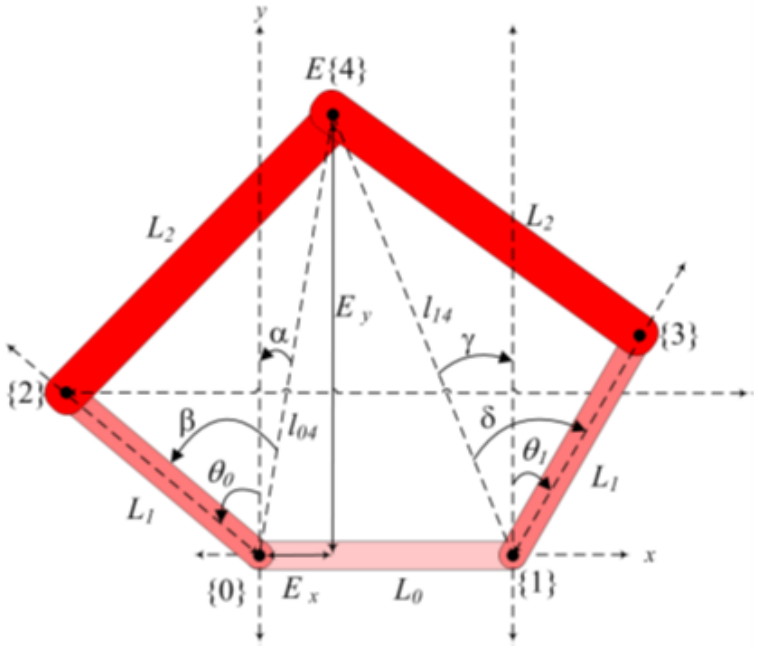


Figure 2: Kinematic diagram with positions P_2, P_3, and midpoint end-effector E

2 System Parameters

Symbol	Description	Value (example)
L_0	Distance between motors	0.076 m
L_1	Length of each motor arm	0.084 m
m_1	Mass of Link 1	0.01479 kg
m_3	Mass of Link 2	0.01918 kg
J_1	Inertia of Link 1	6.51×10^{-7} kg·m^2
J_3	Inertia of Link 2	7.78×10^{-7} kg·m^2
J_m	Rotor inertia	4.0×10^{-6} kg·m^2

Table 1: Model parameters for the simplified Quanser 5-bar system

3 Kinematics

Joint Positions (Manual Convention)

$$P_2 = \begin{bmatrix} -L_1 \sin\theta_0 \\ L_1 \cos\theta_0 \end{bmatrix}, \quad P_3 = \begin{bmatrix} L_1 \sin\theta_1 + L_0 \\ L_1 \cos\theta_1 \end{bmatrix}$$

End-Effector (Midpoint Approximation)

$$E = \frac{P_2 + P_3}{2} \Rightarrow \begin{matrix} E_x = \dfrac{-L_1 \sin\theta_0 + L_1 \sin\theta_1 + L_0}{2} \\ E_y = \dfrac{L_1 \cos\theta_0 + L_1 \cos\theta_1}{2} \end{matrix}$$

Inverse Kinematics (Approximate)

$$\theta_0 = \arcsin\left(\frac{L_1 \sin\theta_1 + L_0 - 2E_x}{L_1}\right)$$
$$\theta_1 = \arcsin\left(\frac{2E_x + L_1 \sin\theta_0 - L_0}{L_1}\right)$$

4 Nonlinear Dynamic Model

Generalized Coordinates

$$q = \begin{bmatrix}\theta_0 \\ \theta_1\end{bmatrix}, \quad \dot{q} = \begin{bmatrix}\omega_0 \\ \omega_1\end{bmatrix}$$

Kinetic Energy

$$T = \frac{1}{2}(J_1 + J_m)\omega_0^2 + \frac{1}{2}(J_3 + J_m)\omega_1^2$$

Lagrangian and Equations of Motion

$$\mathcal{L} = T, \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i}\right) = \tau_i \Rightarrow (J_1 + J_m)\ddot{\theta}_0 = \tau_0, \quad (J_3 + J_m)\ddot{\theta}_1 = \tau_1$$

5 Dynamic Properties of the System

Inertia (Mass) Matrix

The robot dynamics can be expressed in standard form:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta}) + G(\theta) = \tau$$

Here, the inertia matrix is:

$$M(\theta) = \begin{bmatrix} J_1 + J_m & 0 \\ 0 & J_3 + J_m \end{bmatrix}$$

Since there is no coupling between links, this matrix is diagonal.

Gravitational Forces

As the system operates in a horizontal plane, the height of each link's center of mass remains constant. Hence, the gravitational potential energy is constant, and its gradient is zero:

$$G(\theta) = \frac{\partial V}{\partial \theta} = 0$$

Therefore, gravity does not contribute to the dynamics in this configuration.

Damping and Friction

Real joints and motors experience damping and friction. A viscous damping term can be added:

$$C(\dot{\theta}) = \begin{bmatrix} b_1 \dot{\theta}_0 \\ b_2 \dot{\theta}_1 \end{bmatrix}$$

where b_1, b_2 are small positive damping coefficients. In this report, damping is neglected for simplicity but may be included for simulation fidelity or controller tuning.

6 State-Space Model

State and Input Vectors

$$x = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \omega_0 \\ \omega_1 \end{bmatrix}, \quad u = \begin{bmatrix} \tau_0 \\ \tau_1 \end{bmatrix}$$

Dynamics

$$\dot{x} = \begin{bmatrix} \omega_0 \\ \omega_1 \\ \frac{\tau_0}{J_1 + J_m} \\ \frac{\tau_1}{J_3 + J_m} \end{bmatrix}$$

Linearized Model

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{J_1 + J_m} & 0 \\ 0 & \frac{1}{J_3 + J_m} \end{bmatrix}$$

Output: End-Effector Position

From:

$$E_x = \frac{-L_1 \sin\theta_0 + L_1 \sin\theta_1 + L_0}{2}$$
$$E_y = \frac{L_1 \cos\theta_0 + L_1 \cos\theta_1}{2}$$

Jacobian: Output Matrix C

$$\frac{\partial E_x}{\partial \theta_0} = -\frac{L_1}{2}\cos\theta_0, \quad \frac{\partial E_x}{\partial \theta_1} = \frac{L_1}{2}\cos\theta_1$$
$$\frac{\partial E_y}{\partial \theta_0} = -\frac{L_1}{2}\sin\theta_0, \quad \frac{\partial E_y}{\partial \theta_1} = -\frac{L_1}{2}\sin\theta_1$$
$$C = \begin{bmatrix} -\frac{L_1}{2}\cos\theta_0 & \frac{L_1}{2}\cos\theta_1 & 0 & 0 \\ -\frac{L_1}{2}\sin\theta_0 & -\frac{L_1}{2}\sin\theta_1 & 0 & 0 \end{bmatrix}, \quad D = 0$$

7 Conclusion

This report presents a complete model of the Quanser 5-bar planar manipulator using a simplified 2-link approach and the coordinate conventions from the official manual. The model includes full kinematics, a Lagrangian formulation of dynamics, inertial structure, and state-space linearization for control design.